Code Explanation

Configuration Module Explanation

The given code is a configuration module written in Python for the BNCPLS IF23B 10K File Generator. It defines a data class to hold configuration settings for a file generation pipeline.

Overview of the Code

This configuration module is designed to store and manage various settings required for the 10K file generation pipeline. It includes parameters such as SQL source, XSLT files, output directory, file naming conventions, and logging level.

Step 1: Importing Necessary Modules
The code begins by importing necessary modules from Python's standard library. The `dataclasses` module is used to simplify the creation of classes that mainly hold data, and the `typing` module is used for type hints to improve code readability and facilitate static type checking.

```
from dataclasses import dataclass
from typing import Optional
```

Step 2: Defining the PipelineConfig Data Class
The `PipelineConfig` class is defined using the `@dataclass` decorator, which automatically generates special methods like `__init__` and `__repr__` based on the class attributes. This class is designed to hold the configuration for the 10K file generation pipeline.

```
@dataclass
class PipelineConfig:
    """Configuration for the 10K file generation pipeline."""
```

Step 3: Specifying Class Attributes
The `PipelineConfig` class includes several attributes that define the configuration settings. These attributes are categorized into required and optional parameters. Required parameters include `src_sql`, `xslt_file`, `aura15_xslt_file`, and `output_dir`, which are essential for the pipeline's operation. Optional parameters provide default values and can be adjusted according to specific needs.

```
    src_sql: str
    xslt_file: str
    aura15_xslt_file: str
    output_dir: str
    file_prefix: str = "IF23B"
    file_ext: str = "dat"
    batch_id: Optional[str] = None
    run_id: Optional[str] = None
    max_rows_per_file: int = 10000
    log_level: str = "INFO"
```

These steps collectively define a structured and readable way to manage the configuration for the 10K file generation pipeline, making it easier to adjust or extend the configuration as needed.